School of Engineering

E211 – Operations Planning II

Lesson 20 Worksheet

WantTheToys has manufacturing centers located in Singapore, Bangkok and Manila. The company plans to ship a new batch of toys from the manufacturing centers to customers in other ASEAN countries – Jakarta, Phnom Penh, Naypyidaw, Hanoi, Kuala Lumpur and Vientiane. Bandar Seri Begawan has no demand but serves as a transshipment node in the supply chain. The distribution network as well as the supply and demand information at each node is given below.

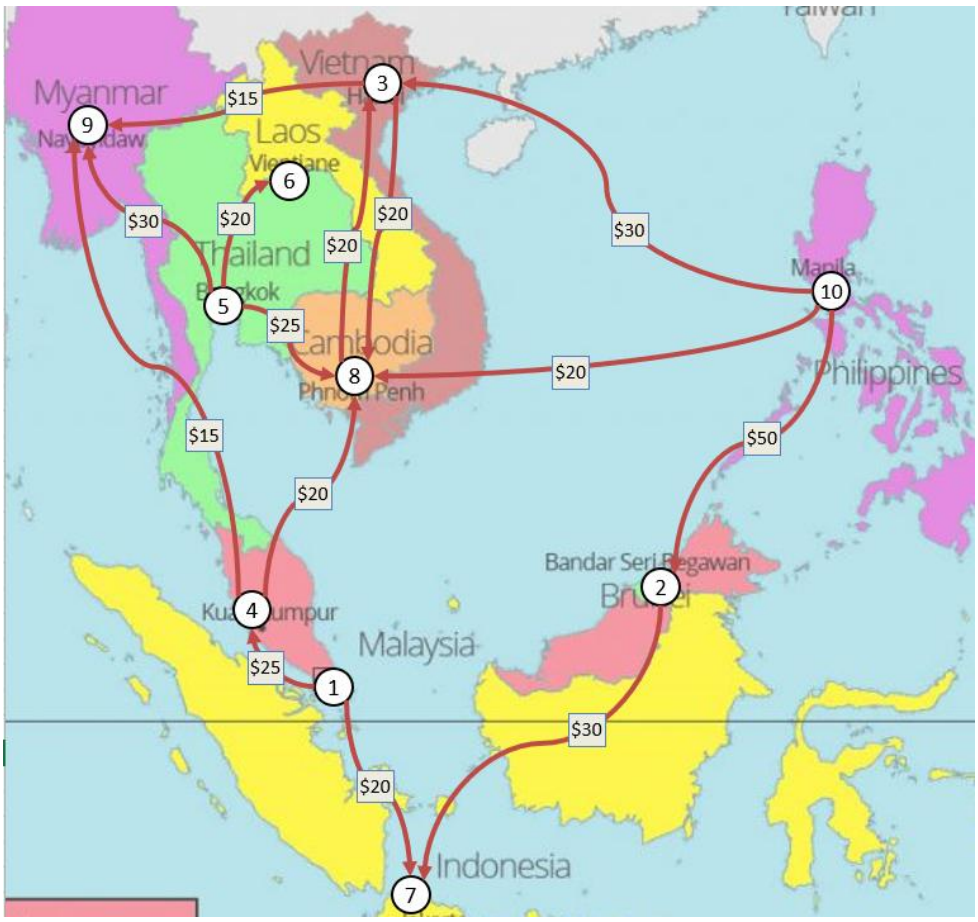


Figure 1: Toy Distribution Network

Table 1: Supply Quantities

Manufacturing Plants	Supply Units
[1] Singapore	**1200**
[5] Bangkok	**800**
[10] Manila	**1000**
Total	**3000**

Table 2: Demand Quantities

Transhipment/ Demand Centres	Demand Units
[2] Bandar Seri Begawan	**0**
[3] Hanoi	**600**
[4] Kuala Lumpur	**800**
[6] Vientiane	**300**
[7] Jakarta	**500**
[8] Phnom Penh	**500**
[9] Naypyidaw	**300**
Total	**3000**

Furthermore, there is a shipping capacity limit of 500 from Manila to Hanoi and 200 from Bangkok to Naypyidaw.

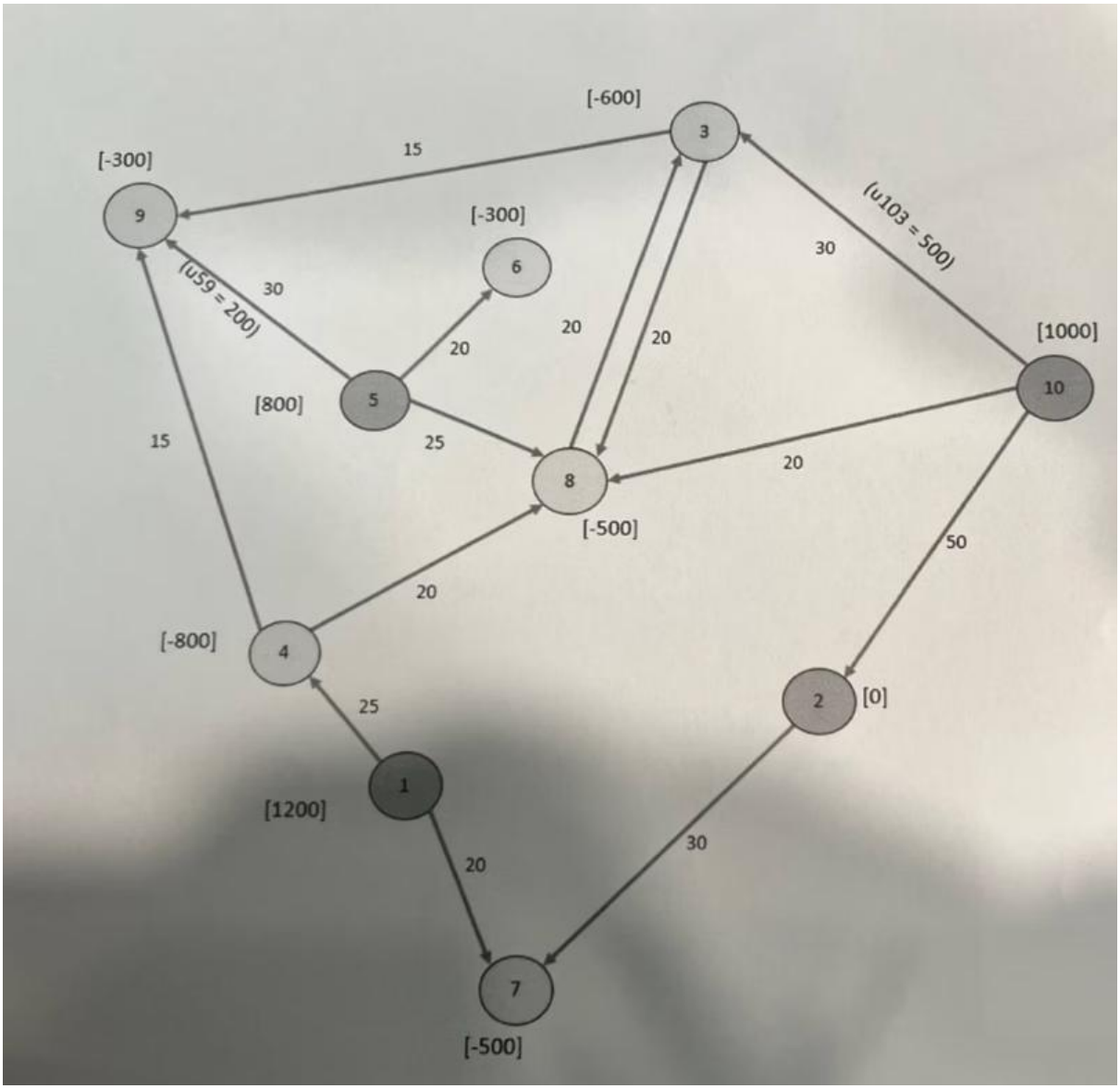
Work out a distribution plan that meets the demand for its educational toys and minimize the total cost of shipment.

1. Based on the information above, complete the table below.

Node	Location	Type of Node (Supply, Transhipment or Demand)	Net flow (number of units; Outflow is +ve, Inflow is -ve)
1	**Singapore**	**Supply**	**1200**
2	Bandar Seri Begawan	Transhipment	0
3	Hanoi	Demand	-600
4	Kuala Lumpur	Demand	-800
5	**Bangkok**	**Supply**	**800**
6	Vientiane	Demand	-300
7	Jakarta	Demand	-500
8	Phnom Penh	Demand	-500
9	Naypyidaw	Demand	-300
10	**Manila**	**Supply**	**1000**

2. What is the function of transhipment? How is Bandar Seri Begawan different from the other nodes?
   Goods are shipped from known supply points (manufacturing centers) to known demand points (customers), possibly through intermediate points (regional and or/field warehouses). It must satisfy the conservation of flow, so that flow in **equals** flow out.

3. Draw a network representation of the routes available to WantTheToys to distribute their toys.



4. The distribution flow problem faced by WantTheToys is known as a **minimum cost flow problem**. Similar to the transportation problem, it has the 'integer solution property' and can be formulated and solved using a linear programming (LP) model.

   a. Define the decision variables for this LP model. In total, how many decision variables are there?

   Decision variables: Xij represents the amount of flow through arc i to j where i, j represents nodes in the network:

   X14, X17, X27, X38, X39, X48, X49, X56, X58, X59, X83, X102, X103, X108

   There are 14 decision variables.

Model answer (MC HoweChiat) from SCORM: Let X_{ij} be the *number of units of toys (in '000s)* to be shipped from node i to node j. In total, there are 14 decision variables:
X_{14}, X_{17}, X_{27}, X_{38}, X_{39}, X_{48}, X_{49}, X_{56}, X_{58}, X_{59}, X_{83}, X_{102}, X_{103}, X_{108}

b. Write down the objective function of the minimum cost flow problem using the decision variables defined above.
Minimize total shipment cost, Z = 25X14 + 20X17 + 30X27 + 20X38 + 15X39 + 20X48 + 15X49 + 20X56 + 25X58 + 30X59 + 20X83 + 50X102 + 30X103 + 20X108

c. There are <u>three types of constraints</u> in the minimum cost flow LP model.

i. Formulate the **Net flow constraint** for each node in the network. This type of constraint has the following form:

Amount shipped out – Amount shipped in = Net flow generated at the node

Node 1: X14 + X17 = 1200
Node 2: X27 – X102 = 0
Node 3: X38 + X39 – X83 – X103 = -600
Node 4: X48 + X49 – X14 = -800
Node 5: X56 + X58 + X59 = 800
Node 6: -X56 = -300
Node 7: -X17 – X27 = -500
Node 8: X83 – X38 – X58 - X108 = -500
Node 9: -X39 – X49 – X59 = -300
Node 10: X102 + X103 + X108 = 1000

ii. Formulate the **Upper-bound constraints** for arcs having limited shipping capacities.
Furthermore, there is a shipping capacity limit of 500 from Manila to Hanoi and 200 from Bangkok to Naypyidaw.
Arc 10 to 3: X103 <= 500
Arc 5 to 9: X59 <= 200

iii. Formulate the **Non-negativity constraint** for each decision variable.
All Xij >= 0

5. Construct the minimum cost flow LP model in Excel Solver (template attached below). Build two models – one without the arc capacity constraints and one with arc capacity constraints. Compare and comment on their solutions.

AY 2025 Sem 2 E211
L20 Template.xlsx

Minimize Total Cost:
Without arc capacity constraints = 80000 (Amt flow = 3100)
With arc capacity constraints = 84000 (Amt flow = 3400)

6. Explain how the constraints in the minimum cost flow model should change for each of the following cases. Obtain a new optimal solution for each case.

   a. The plant in Singapore (Node 1) experienced a disruption and can only supply 1000 units of toys instead of 1200 **(Ans: 79000)**

When Total Supply (2800) < Total Demand (3000),

- This means that not all demand will be fulfilled.
- All demand node sign become “>=” (because RHS value is negative)

Decision Variables	Amount Flow	Unit Cost
X14	600	25
X17	400	20
X27	100	30
X38	0	20
X39	100	15
X48	0	20
X49	0	15
X56	300	20
X58	300	25
X59	200	30
X83	200	20
X102	100	50
X103	500	30
X108	400	20

Minimize Total Cost = 79000

Net Flow Constraints	constraint equation LHS	LHS	Sign	RHS
Supply Node 1:	X14+X17=1000	1000	=	1000
Supply Node 5:	X56+X58+X59=800	800	=	800
Supply Node 10:	X102+X103+X108=1000	1000	=	1000
Transhipment Node 2:	X27-X102=0	0	=	0
Demand Node 3	X38+X39-X83-X103=-600	-600	>=	-600
Demand Node 4	X48+X49-X14=-800	-600	>=	-800
Demand Node 6:	-X56=-300	-300	>=	-300
Demand Node 7:	-X17-X27=-500	-500	>=	-500
Demand Node 8	X83-X38-X58-X108=-500	-500	>=	-500
Demand Node 9:	-X39-X49-X59=-300	-300	>=	-300
Arc Capacity Constraints				
Arc 10-3	X103 <= 500	500	<=	500
Arc 5-9	X59 <= 200	200	<=	200

b. The demand at Phnom Penh is reduced to 400 **(Ans: 81500)**

When Total Supply (3000) > Total Demand (2900),

- This means that not all supply need to be sent out.
- All supply node sign becomes "<="

Decision Variables	Amount Flow	Unit Cost
X14	800	25
X17	400	20
X27	100	30
X38	0	20
X39	100	15
X48	0	20
X49	0	15
X56	300	20
X58	200	25
X59	200	30
X83	200	20
X102	100	50
X103	500	30
X108	400	20

Minimize Total Cost = 81500

Net Flow Constraints	constraint equation LHS	LHS	Sign	RHS
Supply Node 1:	X14+X17=1200	1200	<=	1200
Supply Node 5:	X56+X58+X59=800	700	<=	800
Supply Node 10:	X102+X103+X108=1000	1000	<=	1000
Transhipment Node 2:	X27-X102=0	0	=	0
Demand Node 3	X38+X39-X83-X103=-600	-600	=	-600
Demand Node 4	X48+X49-X14=-800	-800	=	-800
Demand Node 6:	-X56=-300	-300	=	-300
Demand Node 7:	-X17-X27=-500	-500	=	-500
Demand Node 8	X83-X38-X58-X108=-500	-400	=	-400
Demand Node 9:	-X39-X49-X59=-300	-300	=	-300
Arc Capacity Constraints				
Arc 10-3	X103 <= 500	500	<=	500
Arc 5-9	X59 <= 200	200	<=	200